MS - Project Guide

Project Statement:

“Given a **constrained operational budget**, what dispatching and **rebalancing policy** for a fleet of service vehicles **minimizes unmet commuter demand** caused by spatiotemporal imbalances in an urban bike-sharing network?"

Context: In a dock-based bike-sharing system, asymmetric commuting flows between residential, industrial and central zones create severe supply-demand mismatches, resulting in empty stations (origin stockouts) and saturated docks (drop-off rejections). Each station has a fixed maximum bike capacity, so its inventory must remain between 0 and its capacity at all times; any rebalancing operation that loads, unloads, or reroutes bikes must respect these bounds. Under a fixed budget for rebalancing vehicle routes, fuel, and labor, this project investigates vehicle dispatching policies to maximize network service reliability and accessibility.

Questions 1:

- What are the variables of the system?
    - Total fleet size (number of operating bicycles).
    - Total number of stations.
    - City grid (2D matrix defining the location and zones of each city block)
    - Number and capacity of rebalancing vehicles (vans).
    - Rebalancing policy thresholds
    - Incentive schemes (pricing discounts or bonus credits for dropping off bikes at underfilled stations).
    - Number of trips awarded with incentives (to evaluate this scheme)
    - Time of day / Day of week (peak morning commute, evening rush, weekend).
    - Weather conditions (temperature, precipitation, wind).
    - Average user walking.
    - Average waiting time of a user.
    - Max Budget
    - Cost of fuel per block for the vans
    - Cost of charging per minute
    - Number of users that successfully rode a bike from A to B.

- What are the main entities involved?

  Entities: people, station, bicycle, rebalancing van
  Events: bicycle trip, van trip

- What are their properties?

  Entities:
  People - id, origin(x,y), destination(x,y) and wating_time
  Station - id, docks, occupied_docs, position(x, y), zone_type
  Bicycle - id, position, battery
  Rebalancing van- id, position(x,y), max_capacity
  Events:
  Bicycle trip - id, person_id, origin_station_id, destination_station_id, incentive
  Van trip - id, origin_station_id, destination_station_id, nr_bikes

- What are the stochastic processes involved?
  - User Generation (People): Modeled via a Non-Homogeneous Poisson arrival process where spawning rates vary dynamically by time of day (morning vs. evening commute peaks), day of the week, and weather conditions.
  - Trip Destination Selection (Bicycle trip): Probabilistic destination choice sampling `destination_station_id` from a transition matrix conditioned on the destination's `zone_type` (residential, central, industrial) and the current hour.
  - Trip Duration (Bicycle trip & Van trip): Travel times across the city grid follow a right-skewed distribution (e.g., Log-Normal) with random traffic perturbations, determining elapsed trip time, van fuel costs, and accumulated `Person.waiting_time`.
  - User Acceptance of Incentives (People): A Bernoulli decision where a user probabilistically accepts or rejects a pricing discount/bonus credit to reroute their drop-off to an underfilled station.
  - Battery Discharge (Bicycle): Stochastic energy consumption per block traveled based on rider effort, elevation/slope differences, and wind, directly impacting charging time and docking station availability.

- Descriptive, Normative, Predictive, Prescriptive, or Speculative? Why?

  The model is Prescriptive (with a Normative foundation).

  - Prescriptive: It does not merely describe or predict system behavior; it actively determines the best course of action by finding the dynamic van dispatch rules and user incentive policies that optimize system performance (minimizing unmet commuter demand) subject to a constrained operational budget and physical dock capacities.
  - Normative Element: It establishes an objective standard of efficiency and service equity to evaluate how the urban bike-sharing network *ought* to operate.

Questions 2:

- Variables of the System: Exogenous vs. Endogenous

1. Exogenous Variables (External Inputs & Fixed Environment)

These variables originate outside the system boundary; they are imposed on the model and are not affected by bike or van movements.

- **Environmental & Temporal Context:**
  - `Time of day / Day of week` (determines peak/off-peak commute rhythms).
  - `Weather conditions` (temperature, rain, wind modulating overall trip demand and rider willingness).
- **Network & Urban Geography:**
  - `City grid` (the static 2D matrix layout of blocks and their categorical `zone_type`: residential, industrial, central).
  - `Total number of stations` and their fixed dock capacities (`docks`).
- **Economic & Operational Constraints (Resource Limits):**
  - `Max Budget` (the fixed expenditure ceiling for operational activities).
  - `Cost of fuel per block` (van fuel/operational cost rate).
  - `Cost of charging per minute` (electricity/charging unit tariff).
- **Behavioral Thresholds (Baseline User Parameters):**
  - `average_waiting_time` (the global threshold defining how long a user is willing to wait at an empty dock before balking).

2. Endogenous Variables (Internal State, Decisions & System Outputs)

These variables are determined, updated, and produced inside the simulation as a direct result of agent interactions, decisions, and system dynamics.

- **Decision Variables (Policy Controls set by the operator):**
  - `Number and capacity of rebalancing vehicles (vans)`.
  - `Total fleet size` (number of active bicycles deployed).
  - `Rebalancing policy thresholds` (inventory bounds triggering van dispatches).
  - `Incentive schemes` (discount or credit magnitudes offered to riders).
- **Dynamic State Variables (Internal status varying with time t):**
  - `occupied_docks` (bike inventory at each station i at time t).
  - `Bicycle.battery` (individual battery levels discharging and charging).

    - `Rebalancing van position and loaded bikes` (current load and route location).
    - `Person.waiting_time` (accumulated wait duration of waiting agents).
- **Output / Performance Variables (Resulting System KPIs):**
    - `Number of users that successfully rode a bike from A to B` (satisfied demand).
    - `Unmet demand` (count of origin stockouts and destination drop-off rejections).
    - `Average user walking / deviation time` (induced by full/empty stations).
    - `Number of trips awarded with incentives` (user adoption of rerouting discounts).
    - `Total operational expenditure` (accumulated fuel + charging costs).

- Controllable vs. Uncontrollable Variables

1. Controllable Variables (Decision Variables / Levers)

These are parameters directly governed or tuned by the system operator to steer system performance and optimize operations within the budget limit:

- **`Rebalancing policy thresholds`:** The upper/lower inventory boundary triggers that dictate when a station requires van intervention (e.g., dispatch when occupied docks drop below 20% or exceed 80%).
- **`Number and capacity of rebalancing vehicles (vans)`:** The operational fleet size and cargo limit allocated to move bikes between stations.
- **`Incentive schemes`:** The discount rates or bonus credits offered to riders to encourage drop-offs at underfilled docks.
- **`Total fleet size`:** The total quantity of active bicycles injected into the network.
- **`Max Budget` allocation rules:** How the total operational budget ceiling is partitioned between vehicle fuel/maintenance and battery charging operations.

2. Uncontrollable Variables (Exogenous Factors & System State Responses)

These variables are outside the operator's direct real-time control; they represent the operational environment, physical limits, or resulting emergent behaviors:

- **Environmental & Temporal Factors:**

    - `Time of day / Day of week` (dictates peak commuting rush hours vs. off-peak times).
    - `Weather conditions` (rain, wind, and temperature influencing rider willingness).
- **Physical Infrastructure & Fixed Geography:**
    - `City grid` (spatial coordinates and immutable zoning: residential, central, industrial).
    - `Total number of stations` and their maximum physical dock capacities (`docks`).
    - `Cost of fuel per block` and `Cost of charging per minute` (market-driven utility and fuel tariffs).
- **User Behavior & Emerging System States:**
    - `average_waiting_time` (the inherent patience threshold of commuters before balking).
    - `occupied_docks` (real-time station fill levels resulting from decentralized rider trips).
    - `Bicycle.battery` (rate of battery drain during user trips).
    - `Number of users that successfully rode a bike` / `Unmet demand` (resulting service KPIs).

- Questions Answered by the Model (Endogenous Variables to Evaluate)

  The core questions answered by the model directly evaluate the system's **endogenous output variables** (KPIs and performance measures) to assess service quality, financial sustainability, and policy effectiveness:

  ### 1. Service Reliability & Commuter Satisfaction

  - **What proportion of travel demand is successfully served?**
    - *Endogenous Variable:* `Number of users that successfully rode a bike from A to B` / System Service Level (\%).
  - **How severe is unmet demand across the city?**
    - *Endogenous Variable:* Total count of origin stockouts (empty stations) and destination drop-off rejections (full stations).
  - **How much delay and inconvenience do commuters experience due to imbalances?**
    - *Endogenous Variables:* Average user walking deviation distance and accumulated `Person.waiting_time`.

  ### 2. Operational Expenditure & Budget Feasibility

- **Can the rebalancing policy operate strictly within the defined financial budget?**
    - *Endogenous Variable:* Total Operational Expenditure (OPEX), calculated from accumulated van fuel/travel costs and e-bike charging fees.
- **How cost-effective is the vehicle rebalancing routing?**
    - *Endogenous Variable:* Rebalancing efficiency yield (number of bicycles moved per kilometer traveled by the vans).

3. User Incentive Scheme Viability

- **To what extent do crowdsourced user incentives alleviate the logistical burden on vans?**
    - *Endogenous Variable:* `Number of trips awarded with incentives` (rate of commuter participation in self-balancing).
- **Does dynamic discount pricing reduce total operating costs compared to relying exclusively on vehicle dispatches?**
    - *Endogenous Variable:* Net financial cost (incentive payouts vs. saved van fuel and driver hours).

4. Network Equilibrium & Spatial Equity

- **How frequently do stations enter critical failure states throughout the day?**
    - *Endogenous Variable:* Station starvation duration (time spent at 0 bikes) and saturation duration (time spent at full capacity).
- **Is service quality distributed equitably across residential, commercial, and industrial zones?**
    - *Endogenous Variable:* Spatial distribution of unmet demand (e.g., Gini coefficient of station failure rates across different `zone_type` blocks).

Decision Variables (Operational Policies)

The decision variables represent the actionable control levers configured by the system operator to steer system performance, manage rebalancing operations, and allocate the constrained budget:

- **Van Dispatching & Rebalancing Thresholds (\theta_{\text{min}}, \theta_{\text{max}}):**
    - The inventory boundary rules that trigger a service vehicle intervention (e.g., dispatch a van to load bikes when a station exceeds \theta_{\text{max}} = 80\% capacity, or unload bikes when it drops below \theta_{\text{min}} = 20\%).
- **Van Routing & Scheduling Policy:**
    - The operational logic governing vehicle dispatch:

        - *Static / Periodic:* Fixed routes covering pre-assigned station loops at set time intervals.
        - *Dynamic / Reactive:* On-demand dispatching to whichever station reaches a critical starvation/saturation threshold first.
        - *Proactive / Predictive:* Routing based on anticipated short-term commuter demand derived from time-of-day traffic patterns.
- **Vehicle Fleet Allocation (N_{\text{vans}}, C_{\text{van}}):**
    - The number of active rebalancing vans deployed simultaneously and their cargo capacities (bikes per van).
- **Target Inventory Level (S_i^*):**
    - The target fill level (e.g., 50\% of dock capacity) that a rebalancing van aims to restore when servicing station i.
- **User Incentive Policy Parameters:**
    - **Incentive Trigger Bound:** The station occupancy threshold that activates a user reward (e.g., offer incentives when destination docks are >85\% full or origin docks are <15\% full).
    - **Incentive Magnitude / Discount Scheme:** The monetary discount or ride credit amount offered to users to entice them to divert their drop-off or pick-up to an adjacent imbalanced station.
- **Budget Allocation Ratio (\alpha):**
    - The financial split rule dividing `Max Budget` between real-time vehicle operational costs (fuel and driver hours) and battery charging operations.

- Static vs. Dynamic, Deterministic vs. Stochastic, Continuous vs. Discrete

1. Static vs. Dynamic Aspects

- **Static Aspects (Do not change over simulation time):**
    - **Urban Topology & Zoning:** The 2D city grid, block coordinates (x, y), and their assigned zone types (residential, central, industrial).
    - **Station Infrastructure:** Total number of stations, physical locations, and maximum dock capacities (`docks`).
    - **System Parameters & Constraints:** `Max Budget`, unit cost rates (fuel cost per block, charging cost per minute), and van cargo limits (`max_capacity`).
- **Dynamic Aspects (Evolve continuously with time t):**
    - **Station Inventories:** The number of bikes currently docked at each station (`occupied_docks`).
    - **Mobile Agents:** Positions and routes of `People` and `Rebalancing vans`.
    - **Vehicle Payloads & Bike State:** Bikes loaded on vans (`nr_bikes`), bicycle `battery` percentage levels, and docking/charging status.

    - **Accumulated Metrics:** Individual `Person.waiting_time`, budget depletion, and total completed vs. failed trips.
    - **Environmental Time:** Clock progression (`Time of day / Day of week`), driving shift changes and peak/off-peak commuting rhythms.

2. Deterministic vs. Stochastic Aspects

- **Deterministic Aspects (Rule-based, no randomness):**
    - **Physical & Inventory Bounds:** Hard dock capacity limits (bikes docked must strictly remain within [0, \text{docks}]; vans cannot exceed `max_capacity`).
    - **Fixed Economic Accounting:** Fuel and charging cost updates calculated directly from blocks driven and minutes docked:
        - $\Delta Cost = d \times fuel_rate + \Delta t \times charging_rate$
    - **Operational Decision Logic:** Exact threshold triggers for rebalancing (e.g., dispatching when occupancy crosses strictly below 20% or above 80%).
    - **Distance Calculations:** Manhattan or Euclidean grid distance between stations.
- **Stochastic Aspects (Governed by probability distributions):**
    - **Commuter Inflow:** Arrival times and spawning rates of `People` via a Non-Homogeneous Poisson Process conditioned on time and weather.
    - **Origin-Destination Pairing:** Sampling `destination_station_id` from a time-dependent categorical transition matrix.
    - **Transit Duration:** Travel times across the grid featuring random noise representing congestion and cyclist speed variation.
    - **User Decisions:** Probabilistic choice to accept incentive discounts and probabilistic abandonment (*balking*) when stations are empty.
    - **Battery Discharge:** Random variance in battery draw per block traveled due to user effort, incline, or headwind.

3. Continuous vs. Discrete Aspects

- **Discrete Aspects:**
    - **State Quantities:** Countable, integer-valued items (`occupied_docks`, bicycle counts, vans, active users, failed trips).
    - **Time Progression & Scheduling:** Time advances in discrete ticks (e.g., 1 tick = 1 minute) via the Mesa scheduler.
    - **Spatial Units:** City space represented by discrete cells or distinct station nodes on the grid.
    - **State Transitions:** Discrete events (user arrives, trip starts, bike is docked, van unloads 5 bikes).
- **Continuous (or Approximated Continuous) Aspects:**
    - **Financial Expenditure:** Accumulated operational costs and budget balance (real-valued currency).

    - **Battery Degradation & State of Charge:** `Bicycle.battery` tracked as a continuous percentage [0.0, 100.0\%].
    - **Aggregated Ratios:** Service level percentages, station availability rates, and spatial equity metrics (e.g., Gini coefficients).

- Identification of Core Simulation Concepts

1. Agents (Autonomous Decision-Makers & Active Entities)

- People (Commuters / Users): Active decision agents. They observe station inventory, evaluate options, decide whether to accept an incentive, choose whether to wait or balk when encountering empty docks, and walk or reroute if their destination is full.
- Rebalancing Van (Service Fleet): Active operational agents. They execute dispatch commands, navigate the grid to target stations, load/unload batches of bikes, and monitor their own capacity and route schedules.

2. Objects (Passive Entities / Physical Infrastructure)

- Station: Fixed physical docking infrastructure. It holds docks, tracks occupancy, docks/undocks bikes, and charges batteries when bikes are connected.
- Bicycle (E-bike): Mobile physical resource. It maintains position and battery charge, transitions between docked and active states, but moves only when ridden by a person or hauled by a van.
- City Block / Grid Cell: Spatial container holding coordinates (x, y) and terrain/zoning attributes (zone_type).

3. States (Conditions of Entities / System at Time t)

- Station States: Starved / Empty (0 bikes), Nominal / Balanced (between 0 and max capacity), Saturated / Full (at max capacity).
- Person States: Idle / Not Spawned, Waiting at Station, Riding / In-Transit, Rerouting / Walking, Satisfied / Completed, Balked / Abandoned.
- Bicycle States: Docked & Available, Docked & Charging (insufficient battery), In-Use / Ridden, Loaded in Van.
- Van States: Idle at Depot, En Route to Pick-up, Loading Bikes, En Route to Drop-off, Unloading Bikes.
- System States: Total budget remaining, global fleet availability, and current network-wide service level.

4. Events (Instantaneous State Transitions)

- User Arrival: A person spawns at an origin station.

- Trip Start (Checkout): A user unlocks an available bike; station inventory decreases by 1, and user state transitions to Riding.
- Origin Stockout Event: A user arrives at an empty station (0 bikes).
- Balk Event: A user waiting time exceeds their patience threshold and they abandon the system.
- Incentive Offered / Accepted Event: A rerouting discount is generated and agreed upon by the rider.
- Trip End (Return): A user docks a bike at a destination station; station inventory increases by 1.
- Destination Saturation Event: A rider reaches a full station and cannot dock.
- Dispatch Trigger Event: Station inventory crosses a critical threshold (upper or lower bound), initiating a van dispatch order.

5. Activities (State Durations with Explicit Start and End)

- Commuter Riding: The active travel time of a user cycling from origin to destination across the grid.
- User Waiting: The elapsed duration a commuter spends waiting at a depleted station hoping for a bike.
- Van Transit: The driving time required for a service vehicle to travel between stations across grid blocks.
- Bike Servicing / Handling: The physical duration required for a van operator to load or unload bikes at a docking station.
- Dock Charging: The duration a depleted e-bike remains connected to a dock drawing power to reach full charge.

6. Processes (End-to-End Sequences of Events and Activities)

- Commuter Trip Process:
  1. Arrival Event
  2. Unlock Event
  3. Riding Activity
  4. Docking Event
  5. User Exit
- Balking / Rerouting Process:
  1. Arrival Event
  2. Stockout Event
  3. Waiting Activity
  4. Patience Exceeded Event
  5. Walking / Abandonment
- Vehicle Rebalancing Process:
  1.Threshold Violation Event
  2. Van Dispatch Event
  3.Transit Activity (to origin station)
  4. Loading Activity
  5. Transit Activity (to destination station)
  6. Unloading Activity

- Battery Lifecycle Process:
  1. Discharge during Riding Activity
  2. Dock Connection Event
  3. Recharge Activity
  4. Full Charge Event

Questions 3:

- Restate the goals of your projects according to your own understanding

  The core goal is to determine the optimal operational policy that minimizes unmet commuter demand (empty origins and full destinations) in an urban bike-sharing network subject to a constrained operational budget.

  Specifically, this involves:

  - **Evaluating Van Policies:** Comparing reactive threshold-based dispatching against proactive routing to prevent station stockouts and dock saturation.
  - **Assessing Crowdsourced Incentives:** Determining whether offering user discounts or credits can effectively substitute for costly van trips through self-rebalancing.
  - **Optimizing Budget Allocation:** Balancing the financial trade-off between vehicle routing costs (fuel and labor) and e-bike charging expenses under fixed budget constraints.
  - **Capturing Spatiotemporal Asymmetries:** Simulating how directional commuter flows between residential, central, and industrial zones trigger network imbalances during peak hours.

- Organise the conceptual model and the main logical models

  ### 1. Conceptual Model

  The conceptual model abstracts the real-world bike-sharing system into essential components, assumptions, interactions, and performance metrics according to the modeling lifecycle:

  - **Assumed System & Domain:** An urban dock-based e-bike sharing system across heterogeneous urban zones (residential, commercial/central, industrial). Commuters check an application to find an available bike within walking distance, ride to their target zone, and return it to a dock. Imbalances caused by one-way commuter flows are addressed via physical van

rebalancing and crowdsourced user incentives under an operational budget ceiling.

- **Entities (Dynamic objects that move, change status, and affect other entities):**
  - **People:** Commuters who spawn at an origin coordinate, query the system app, walk to an eligible origin station, ride to their destination, or balk if no bikes are within range.
  - **Rebalancing Vans:** Service vehicles that navigate the city grid to load bikes from saturated stations and unload them at starved stations.
- **Attributes (Local characteristics specific to each entity):**
  - **People:** `id`, `origin` coordinates (x, y), `destination` coordinates (x, y), `assigned_origin_station_id`, `waiting_time`, and `walking_distance`.
  - **Rebalancing Vans:** `id`, position coordinates (x, y), `max_capacity`, and current bike load (`nr_bikes`).
- **Resources (System elements that entities seize, use, and release):**
  - **Bicycles (E-bikes):** Seized by a commuter at the origin station, used during transit, and released at the destination dock.
  - **Station Docks:** Physical slots holding bicycles; seized when a bike docks or is unloaded by a van, and released when a bike is unlocked or loaded into a van.
- **Variables (Global metrics reflecting the state of the whole system):**
  - `occupied_docks` across all stations (network inventory distribution).
  - `remaining_budget` (funds available for fuel, labor, and battery charging).
  - `current_time` and `weather` (exogenous variables driving peak/off-peak arrival rates).
- **Events (Instantaneous occurrences changing system state):**
  - `Commuter Spawn Event`: User appears at spatial origin (x, y).
  - `App Query & Station Selection Event`: The app evaluates nearby stations and assigns an optimal station with available bikes.
  - `App Balk Event`: No stations within walking range have bikes; commuter immediately abandons the trip.
  - `Trip Start (Unlock) Event`: User arrives at dock and unseizes bike.
  - `Trip End (Return) Event`: User docks bike at destination station.
  - `Van Dispatch Event`: Station occupancy crosses an inventory threshold (\theta_{\text{min}} or \theta_{\text{max}}).
- **Activities (State durations with specified length):**
  - `Access Walking Activity`: Time spent by the commuter walking from spawn origin (x, y) to the assigned station.
  - `Commuter Riding Activity`: Time duration spent cycling from origin dock to destination dock.
  - `Van Transit Activity`: Driving duration between stations across grid blocks.

    - `Bike Handling Activity`: Time taken by a van operator to load or unload bicycles at a dock.
    - `Dock Charging Activity`: Duration an e-bike remains connected to a dock drawing power to reach operational battery charge.
- **Data Requirements & Performance Measures:**
    - **Data:** Station coordinates, dock capacities, hourly Poisson arrival rates, origin-destination zone transition matrices, walking speed, unit fuel and charging costs.
    - **Performance Measures (KPIs):** Unmet commuter demand (app balks and destination rejections), completed trips, average access walking distance, station starvation/saturation durations, and total operational cost.

2. Main Logical Models

The logical models establish the operational algorithms, decision logic, and flowcharts governing state transitions across the simulation:

Logical Model 1: Commuter Generation, App Selection & Trip Flowchart

1. **User Spawn:** A commuter is generated at coordinate (x_{\text{origin}}, y_{\text{origin}}) via a Non-Homogeneous Poisson arrival process conditioned on the hour and weather.
2. **App Evaluation (Candidate Set Filtering):**
    - The system evaluates candidate stations s_k satisfying:
    - $$dist(origin, s_k) \leq R_{walk} and B_{s_k}(t) > 0$$
    - where R_{\text{walk}} is the maximum allowable walking radius and B_{s_k}(t) is the count of available, charged bikes.
3. **Branch Decision:**
    - **If Candidate Set is Empty:** Commuter logs an `App Balk Event` (unmet demand due to local starvation) and immediately leaves the system without walking.
    - **If Candidate Set is Non-Empty:**
        - Select optimal station s^* minimizing generalized travel cost: $$s^* = \arg\min_{s_k} \Big(w_{\text{walk}} \cdot \text{dist}(\text{origin}, s_k) + \text{dist}(s_k, \text{dest}) \Big)$$
        - Assign `origin_station_id = s*` and initiate `Access Walking Activity` for duration t = \text{dist}(\text{origin}, s^*) / v_{\text{walk}}.
4. **Dock Arrival & Unlock:** Commuter arrives at s^*. If inventory is intact, commuter seizes bike, releases dock, and begins `Commuter Riding Activity`.
5. **Incentive Evaluation & Return:**

    - Near destination s_{\text{dest}}, if s_{\text{dest}} is full or near capacity and an adjacent station s_{\text{alt}} offers an incentive discount, the commuter probabilistically decides via Bernoulli trial whether to divert.
    - Commuter reaches destination dock:
        - *If dock available:* Seize dock, release bike, record battery state, exit system as satisfied.
        - *If station full:* Commuter diverts to the nearest dock with open capacity, logging detour walking distance.

Logical Model 2: Van Dispatching & Physical Rebalancing Flowchart

1. **Threshold Assessment:** At each time step, check station occupancy ratio B_i(t) / C_i against operational thresholds:
    - Flag as *Deficit / Starvation Target* if B_i(t) < \theta_{\text{min}}.
    - Flag as *Surplus / Saturation Target* if B_i(t) > \theta_{\text{max}}.
2. **Budget Verification:** Check whether accumulated expenses are below `Max Budget`.
    - *If Budget Depleted:* Abort dispatch; vans remain idle at the depot.
    - *If Budget Available:* Proceed with route dispatch.
3. **Route & Service Execution:**
    - Van calculates the shortest grid path to the target station and executes `Van Transit Activity`, deducting fuel cost per block driven.
    - Upon arrival, van executes `Bike Handling Activity` (loading bikes at surplus stations or unloading at deficit stations) to restore station inventory toward target level S_i^*.
    - Dock and bike resources are updated in discrete steps within vehicle capacity bounds.

Logical Model 3: Battery Degradation & Charging Flowchart

1. **Discharge on Trip:** During riding, battery level decreases stochastically based on distance, rider effort, and wind.
2. **Dock Connection:** Upon return, if battery charge falls below operational threshold, the bicycle transitions into `Dock Charging Activity`.
3. **Resource State:** The bike remains unavailable for subsequent commuter app queries until it reaches minimum operational charge, accumulating charging expenses against the operational budget per minute.

- Define your data requirements (real-world datasets, synthetic data, random data)

1. Real-World Datasets

These provide the empirical spatial and structural grounding for Porto:

- **Station Network & Dock Capacities:** Spatial coordinates (x, y) of existing bike stations and their physical dock limits (e.g., from Porto's public bike-sharing system or open municipal GIS / OpenStreetMap data).
- **Zoning & Land Use:** Classification of city blocks into categorical zone types (residential, central business district/commercial, university/industrial) to anchor commuter origins and destinations.
- **Economic & Operational Costs:** Real-world benchmark unit costs (local commercial electricity tariffs for e-bike charging per minute, van diesel/fuel consumption cost per kilometer, driver labor rates).

2. Synthetic Data

Generated deterministically or semi-empirically based on realistic commuter assumptions to bridge gaps where full empirical origin-destination traces are unavailable:

- **Origin-Destination (O-D) Transition Matrices:** Pre-computed probability matrices mapped to city zones and hours (e.g., in the morning peak, 75% of trips originate in residential zones and terminate in central/university zones; inverted in the evening).
- **Distance Matrix:** Pre-calculated grid/Manhattan distance matrix between all pairs of stations on the grid.
- **Topographical Slope Map:** Elevation values assigned to grid blocks to calculate slope variations for bike battery drain.

3. Random / Stochastic Data (Sampled During Runtime)

Variables sampled dynamically at each time step using fitted probability distributions:

- **Commuter Arrivals:** Hourly arrival times sampled from a Non-Homogeneous Poisson process conditioned on time of day, day of week, and weather.
- **Trip Durations:** Travel times generated from continuous right-skewed distributions (e.g., Log-Normal or Weibull) to capture traffic noise, red lights, and varying pedaling speeds.
- **Individual Commuter Attributes:**
    - Exact continuous spawn coordinate (x, y) within a zone.
    - Individual walking tolerance radius (R_{\text{walk}}) sampled around the system average.
    - Incentive acceptance decisions sampled via a Bernoulli distribution.
- **Battery Discharge Perturbations:** Gaussian noise added to base energy consumption per block to model variable rider weights and pedaling assist levels.

Extra bullet points:

- Van rebalancing policies and User incentive schemes

1. Van Rebalancing Policies to Evaluate

In the simulation experiments, several van dispatching policies are compared against a baseline to determine the most cost-effective operational strategy:

- Policy 0: No Rebalancing (Baseline):
  Vans remain completely inactive at the depot.
  Used as a benchmark to measure natural system entropy, uncontrolled station starvation/saturation rates, and baseline unmet demand.
- Policy 1: Static / Periodic Route (Fixed Schedule):
  Vans patrol a fixed cyclical loop across pre-assigned station routes at regular time intervals (e.g., every 60 minutes).
  At each stop, the driver rebalances station inventory toward a static target (50% capacity) regardless of real-time demand.
  Characteristics: Low computational overhead, predictable fuel usage, but slow to respond to localized rush-hour spikes.
- Policy 2: Reactive Threshold-Based Dispatching (Static Bounds):
  Triggered whenever any station inventory crosses fixed lower or upper safety thresholds:
  Unload trigger: Occupancy drops below 20% capacity.
  Load trigger: Occupancy exceeds 80% capacity.
  A van is dispatched greedily to the nearest violated station to move bikes to/from the closest complementary station (surplus to deficit).
  Characteristics: Responsive to emergent stockouts, but risks high fuel expenditure due to erratic backtracking across the grid.
- Policy 3: Proactive / Dynamic Demand-Driven Dispatching:
  Vans are dispatched based on anticipated stockouts over a short forward-looking time horizon (e.g., predicting net deficit 20–30 minutes ahead based on time-of-day Poisson arrival rates and zone commute vectors).
  Vans prioritize high-throughput commercial and transit hub stations before the morning/evening peak arrives.
  Characteristics: Prevents starvation before it happens, resulting in the highest service level, but requires algorithmic forecasting.

2. User Incentive Schemes to Evaluate

User incentives crowdsource rebalancing by rewarding flexible commuters to modify their drop-off station, borrowing concepts from GIRA in Lisbon (which historically used a gamified points-and-bonus credit system):

- Scheme 0: No Incentives (Operator-Only Rebalancing):
  Users always pick up at their closest station with bikes and drop off at their exact destination station.
  Van operations bear 100% of the rebalancing burden.

- Scheme 1: Static Flat Credit / Ride Discount:
  If a commuter returns a bike to a designated "starved" station (occupancy < 20%) or picks up from a "saturated" station (occupancy > 80%), they receive a fixed reward (e.g., 0.20€ ride credit or free extra trip time).
  Acceptance depends on a Bernoulli trial comparing the fixed incentive to the added walking distance from the original destination.
- Scheme 2: Dynamic Distance-Weighted Bonus:
  The incentive reward scales dynamically based on the degree of station imbalance and the detour walking distance required:

$$Incentive = \beta_0 + \beta_1 \cdot DetourDistance + \beta_2 \cdot (TargetDeficit)$$

- Highly congested stations offer higher credits (up to 0.50€ – 0.80€ per trip) during peak commute windows to aggressively incentivize diversion.
- Scheme 3: Hybrid Van-Incentive Coordination:
  Incentives are activated selectively in peripheral or low-density zones where dispatching a physical van is cost-prohibitive due to driving distance.

  Physical vans are reserved exclusively for high-volume arterial corridors.

- Costs, Financial Constraints and Economic Validation

Here is the dedicated **Costs, Financial Constraints, and Economic Validation** section formatted in clean plain text, with fully referenced real-world benchmarks (including EMEL's GIRA in Lisbon and Portuguese utility standards) ready to paste into Google Docs.

3. Operational Costs, Budget Constraints, and Empirical Justification

A. Direct Operational Cost Parameters (OPEX)

- **Van Routing and Transit Cost (c_van = 0.30 EUR / block or ~25.00 EUR / operational hour)**
  - Physical Meaning: Combines vehicle fuel/maintenance expenses, capital leasing amortization, and driver labor allocation per unit of distance/time.
  - Empirical Grounding and Verification:
    1. Driver Labor (Portugal Municipal Standard): A professional light-commercial driver in Portugal has a gross base wage of approximately 1,150.00 to 1,300.00 EUR/month. Under Portuguese labor legislation, the total cost to the municipal operator (EMEL/Porto equivalents) must incorporate statutory employer contributions (TSU -

Taxa Social Unica of 23.75%), mandatory occupational accident insurance (1.5%), daily meal subsidies (subsídio de refeição at ~6.00 EUR/day exempt limit), and mandatory 13th/14th month allowances (subsídio de férias e de Natal). This yields a real employer labor cost of 12.00 to 13.50 EUR per productive working hour.

2. Fuel & Vehicle Operation: Operating a standard urban service van (e.g., Renault Master / Mercedes-Benz Sprinter) in urban stop-and-go conditions results in real-world fuel consumption of 9.0 to 10.5 L/100 km. At Portuguese commercial diesel pricing (~1.60 EUR/L), pure fuel accounts for ~0.16 EUR/km. Adding commercial vehicle leasing depreciation, tire wear, and routine brake/chassis servicing adds ~10.00 to 11.50 EUR per vehicle operating hour.
3. Benchmark Validation: The composite operational hourly rate equals ~24.50 to 25.00 EUR/hour (approx. 0.42 EUR/minute). On an urban grid where transit speed with traffic and intersections averages 15 to 20 km/h (covering roughly 1.4 blocks per minute), this translates reliably to 0.30 EUR per city grid block.

- **Bike Handling & Loading Cost (c_handling = 0.12 EUR / bike)**
  - Physical Meaning: Marginal labor cost incurred whenever a service driver physically latches, unlatches, checks, or secures an individual bicycle into/out of the van and dock.
  - Empirical Grounding and Verification:
    1. Standard logistics time-and-motion studies for micromobility operations establish that an operator spends between 30 and 45 seconds per bicycle to release the dock lock mechanism, inspect the physical vehicle, and fasten it into the van cargo rack (or vice versa).
    2. At the calibrated municipal driver cost of ~12.80 EUR/hour (~0.213 EUR/minute), 35 seconds of physical handling time represents exactly: (35 / 60) * 0.213 EUR = 0.124 EUR per bicycle handled.
- **E-Bike Dock Charging Cost (c_charge = 0.006 EUR / active charging minute)**
  - Physical Meaning: Cost of electricity consumed plus dock charging port maintenance during battery replenishment.
  - Empirical Grounding and Verification:
    1. Battery Specs: Pedelec batteries used in public fleets (such as GIRA in Lisbon) typically feature 36V packs with capacities between 400 Wh and 500 Wh (0.40 to 0.50 kWh) and take 2.5 to 3.5 hours for a nominal charge cycle (20% to 100%).
    2. Energy Tariffs: Regulated commercial low-voltage electricity tariffs in Portugal (published by ERSE - Entidade Reguladora dos Serviços Energéticos) average 0.20 to 0.24 EUR/kWh during standard operating hours.
    3. Unit Calculation: Drawing 0.45 kWh over 180 minutes of dock charging draws an energy cost of (0.45 kWh * 0.22 EUR/kWh) = ~0.10 EUR per full battery charge (~0.00055 EUR/minute in pure electricity). Including smart-dock electronic wear, physical charging pin degradation, and maintenance amortization brings the realistic operational rate to 0.006 EUR per active charging minute (~0.36 EUR/hour).

- **User Rebalancing Incentive Reward (c_incentive = 0.25 EUR / successfully diverted trip)**
    - Physical Meaning: Monetary credit, fee discount, or loyalty value granted to commuters who self-rebalance the system by taking from full stations or returning to starved stations.
    - Empirical Grounding and Verification:
        1. The Real GIRA Benchmark: EMEL implemented this mechanism directly in Lisbon through the GIRA Pontos system (documented in GIRA Terms & Conditions, Clause 6):
            - Users earned 100 bonus points for starting a trip at a station with >70% occupancy.
            - Users earned 100 bonus points for ending a trip at a station with <30% occupancy.
            - Points were convertible into free daily access passes (standard value 2.00 EUR) or waiver of standard trip fees (a 45-minute casual trip is valued at 0.20 to 0.50 EUR). A reward equivalent to 0.25 EUR directly matches GIRA's real bonus structure.
        2. Economic Arbitrage: Moving a bicycle using a van costs between 1.50 and 3.00 EUR (factoring travel distance, fuel, and handling labor). Paying a commuter 0.25 EUR to move that same bicycle achieves an 80% to 90% cost savings for the municipal operator, providing a fully justified financial return.

B. Fleet Sizing & Budget Constraints

- **Reference Urban Network Calibration (Porto Central Core Area):**
    - Stations: 25 to 30 docking stations distributed across residential, central business, and university zones.
    - Active Fleet: 250 electric bicycles (~8 to 10 bikes per station).
    - Operational Shift: 12-hour active daytime window (07:00 to 19:00).
    - Dedicated Vehicles: 2 rebalancing vans (capacity of 15 to 20 bicycles each).
- **Shift Operating Budget (B_max = 180.00 EUR per 12-hour simulation cycle):**
    - Real-World Basis: In European public micromobility operations, dedicated rebalancing operations are allocated between 20% and 30% of total system operational expenditure. Running 2 vans continuously at 100% capacity for 12 hours would cost: 2 vans * 12 hours * 25.00 EUR/hour = 600.00 EUR/day.
    - A municipal public operator cannot subsidize continuous non-stop van mileage. Allocating a constrained budget ceiling of 180.00 EUR per 12-hour cycle restricts physical van utilization to a realistic 30% duty cycle, forcing the optimization algorithm to dispatch vans only when critical bottlenecks occur and to leverage crowdsourced user incentives.
- **Dynamic Budget Enforcement Rule:**
    - Accumulated expenditure is tracked at each discrete simulation step: Total OPEX = Fuel/Van Costs + Handling Costs + Charging Costs + Incentives Paid
    - Once Total OPEX reaches B_max:
        1. Active van dispatches are suspended; vans return to depot (van travel costs drop to zero).

2. Financial incentives for users are deactivated.
3. The network must run solely on natural commuter circulation and passive dock charging for the remainder of the period.

C. Performance Evaluation Metric (Economic Return on Investment)

To evaluate which policy achieves the best trade-off between expenditure and service quality, the simulation computes the Cost per Prevented Stockout:

- Cost Efficiency = Total Budget Spent (EUR) / (Baseline Unmet Demand - Policy Unmet Demand)
- Optimal Policy Criterion: The rebalancing strategy that minimizes unmet commuter trips per euro spent while strictly honoring the B_max <= 180.00 EUR constraint.

- Some implementation notes to consider
  - **Routing and Infrastructure Layer (Mesa Grid):**
    - The spatial environment is implemented using Mesa's grid space (such as `MultiGrid` or `ContinuousSpace`), structured around city blocks.
    - Bicycles and vans must strictly move along the roads that border the blocks, rather than cutting diagonally across block interiors. Movement is constrained to road coordinates, advancing block-by-block toward destination stations.
  - **Porto-Calibrated Spatial Network & Station Placement:**
    - The spatial grid and docking station placement must mirror the real-world geography and urban morphology of Porto:
      - Zone Classification: Distinct, realistic zoning must be assigned across the network, separating residential neighborhoods (e.g., Paranhos, Ramalde), central business/commercial areas (e.g., Baixa/Cedofeita), and university/industrial poles (e.g., Asprela).
      - Station Siting: Docking stations must be placed realistically based on these zone typologies, with high-capacity stations anchored at major intermodal transit hubs (Metro do Porto stations such as Trindade, São Bento, Casa da Música, and Pólo Universitário) to accurately model first-mile/last-mile commuter flows.
  - **Interactive Visualization Dashboard (Mesa / SolaraViz):**
    - Dynamic Real-Time Visualization: The frontend must render the entire system in motion, visually tracking the live positions and states of all active agents (commuters walking/riding along road segments, docking stations updating fill levels, and rebalancing vans in transit).
    - Temporal Controls: Include an interactive simulation speed slider to adjust step frequency (ticks per second), alongside a clear real-time clock display showing the current simulation hour, day phase, and peak window.

- Interactive Policy Configuration: Provide real-time UI controls to dynamically switch and test operational strategies during or between runs, including van dispatch policies (none, static route, threshold-based, proactive) and active user incentive schemes.